

Classifying Constructability of Regular n -gons using Algebra.

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1 Introduction

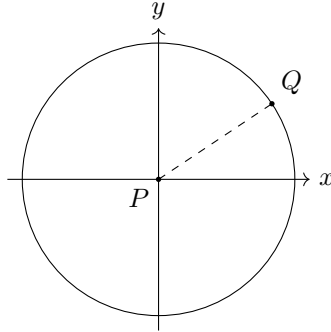
In this paper, I seek to prove that a regular n -gon is constructible if and only if $n = 2^k p_1 p_2 \dots p_t$ where the p_i 's are distinct Fermat primes, and $k \in \mathbb{N} \cup \{0\}$.

2 Constructability of Regular n -gons

Suppose we are playing a game where we are given two points, P and Q , a compass, and a straightedge. Given just these objects, it may seem obvious that we can construct the line $\overline{PQ} = l(P, Q)$.



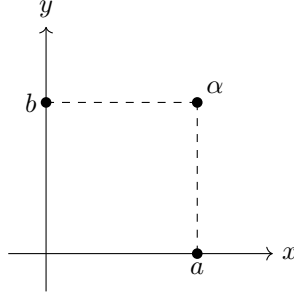
In addition, we can construct a circle with center P and radius $|PQ|$. We'll call this circle $C(P, Q)$.



So now we will start with two points in the Euclidean plane. Without loss of generality, consider these points $0, 1 \in \mathbb{C}$. We can define some other points as follows:

Definition 1. A point is **constructible** if its the intersection of lines and/or circles as formed above. A length l is **constructible** if there exists constructible points P, Q such that $|PQ| = l$. An angle θ is **constructible** if there exists constructible points P, Q, R with $\angle PQR = \theta$.

Remark 1. We may realize that a point α is constructible $\iff \alpha = a + bi$ for a, b constructible numbers:



Claim 1. If α, β are constructible points:

- $\alpha + \beta$ is constructible,
- $|\alpha|$ is constructible,
- $n\alpha$ is constructible for all n .

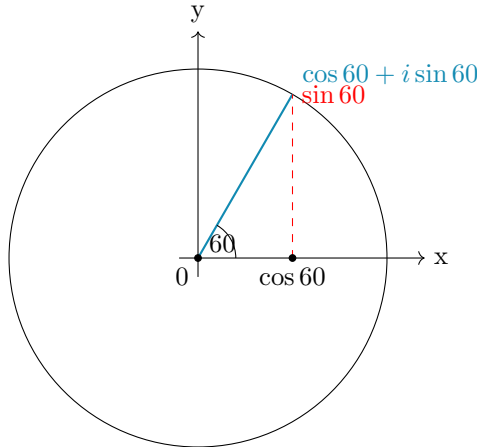
Proof. Assume α is the point on the x -axis at coordinate $(\alpha, 0)$ and β is a point at $(\beta, 0)$. We want to construct $(\alpha + \beta, 0)$. Take the point at 1 as a "unit length". If α is constructible, we can draw a segment of length α starting from 0, and similarly, we can draw a segment of length β . Place the segment of length β starting at the point α , and the end point of this segment of length β starting at α will be exactly at $\alpha + \beta$. ✓

If $\alpha \geq 0$, then we trivially construct $|\alpha|$. If $\alpha < 0$, then we can just show that $-\alpha$ is constructible. We do this by drawing a line through 0 and α . Then we draw a circle centered at 0 with radius equal to that distance. The intersection of the circle with the x -axis is at $-\alpha$. ✓

If α is constructible, we can draw a segment of length α . To get 2α , we can simply place another segment of length α end to end along the same line. For 3α , we place a third segment of length α . So $n\alpha$ is just the sum of α repeated n times. ✓ □

Remark 2. For $r > 0$, we have r is constructible as a length $\iff r$ is constructible as a point.

Remark 3. An angle θ is constructible $\iff \cos\theta + i\sin\theta$ is constructible $\iff \cos\theta$ is constructible.



Lemma 1. Let r, s be constructible points. Then $rs, \frac{r}{s}$ is constructible.

Proof. Let r and $s \neq 0$ be constructible numbers. Draw a line segment of length r (from 0 to r). Then draw a separate line segment of length 1. Using a compass, construct a triangle similar to a reference triangle where the sides satisfy the proportion:

$$\frac{rs}{1} = \frac{r}{1} \cdot s \quad (1)$$

Realize that the corresponding side of the new triangle has length rs . ✓

Similarly, to construct $\frac{r}{s}$,

$$\frac{r}{s} = x \implies x : 1 = r : s \quad (2)$$

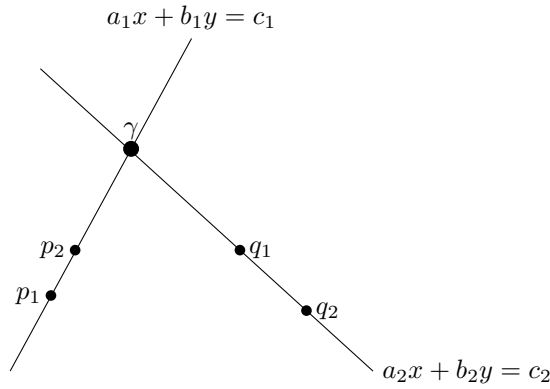
So construct a triangle with sides in proportion $r : s = x : 1$. The side length $x = \frac{r}{s}$ is constructible. ✓ □

Corollary 1. *Let α, β be constructible points. Then $\alpha\beta$ and α/β are constructible points.*

Corollary 2. *The set of constructible points is a subfield of $\mathbb{C}$. Also, $\{\pm \text{constructible lengths}\}$ is a subfield of $\mathbb{R}$.*

So the natural question emerges of which field extensions are generated by constructible points. Suppose we have $F \subseteq \mathbb{C}$ is a constructible field, and we wish to add a new point. We have three methods to do this:

1. Intersect two lines:

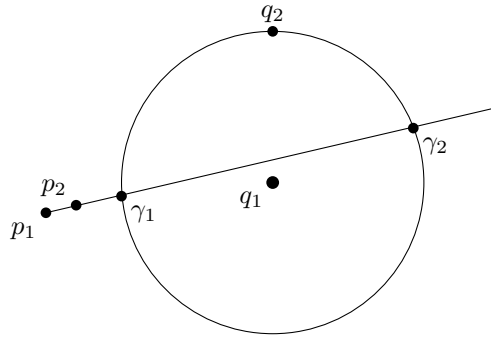


Then γ is the solution to:

$$\begin{aligned} a_1x + b_1y &= c_1 \\ a_2x + b_2y &= c_2 \end{aligned} \quad (3)$$

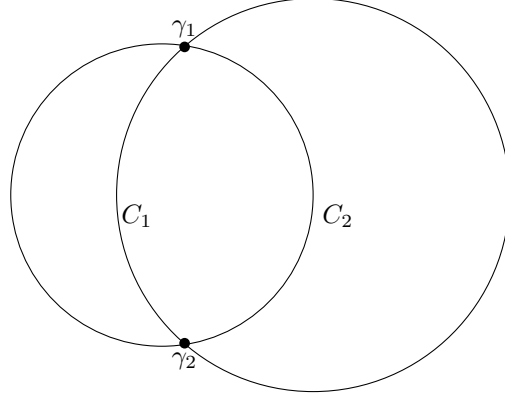
This implies the components of γ lie in a degree 1 extension of F , which implies $\gamma \in F$.

2. Intersect a line and a circle:



So we have that the components of γ_i lie in a degree 2 extension of F by the fact that the equation of the circle is $(x - x_0)^2 + (y - y_0)^2 = r^2$.

3. Intersect two circles:



So we have $(x - x_1)^2 + (y - y_1)^2 = r_1^2$ and $(x - x_2)^2 + (y - y_2)^2 = r_2^2$. This is also a quadratic extension.

Theorem 3. *A constructible point has components which lie in a finite tower of extensions of $\mathbb{Q}$ of the form $\mathbb{Q} \subsetneq \mathbb{Q}(\sqrt{a_1}) \subsetneq \mathbb{Q}(\sqrt{a_1}, \sqrt{a_2}) \subsetneq \cdots \subsetneq \mathbb{Q}(\sqrt{a_1}, \dots, \sqrt{a_n}) = F$. Note that $[F : \mathbb{Q}] = 2^n$*

Proof. So we know that the only new points we can get are in the form:

- intersections of two lines
- intersections of a line and a circle
- intersections of two circles

Define E as the field generated over $\mathbb{Q}$ by the coordinates of the currently available points. We can show that the coordinates of any new intersection point lie either in E , or a quadratic extension $E(\sqrt{\delta})$ for some $\delta \in E$. We will consider cases:

1. **Two lines:** A line has equation $ax + by + c = 0$, with $a, b, c \in E$. Solving two linear equations gives a unique (x, y) with $x, y \in E$. So intersections of two lines do not actually extend the field in the first place.
2. **Line and circle:** A circle with center $(h, k) \in E^2$ and radius $r \in E$ has equation $(x - h)^2 + (y - k)^2 = r^2$. If we intersect such a line in the form $ax + by + c = 0$, it can reduce to substituting a linear expression for y into the circle equation and obtaining a quadratic equation with coefficients in E . So we have that each coordinate of the intersection point is a root of a quadratic polynomial with coefficients in E . If the quadratic is reducible over E , then the roots lie in E . Otherwise, they lie in a quadratic extension $E(\sqrt{\delta})$.
3. **Two circles:** We may subtract the two circle equations to obtain a linear equation in x, y with coefficients in E . Then we can substitute the relation back into one circle to get a quadratic equation over E . Then we are left with a similar situation as above, where either the coordinates lie in E or in a quadratic extension $E(\sqrt{\delta})$.

To build a tower, start with $E_0 = \mathbb{Q}$. Perform constructions one at a time. If the i -th step needs to adjoin a square root $\sqrt{\delta_i}$, we can define $E_i = E_{i-1}(\sqrt{\delta_i})$. If the new point already has coordinates in E_{i-1} , then $E_i = E_{i-1}$. So we have a tower:

$$\mathbb{Q} = E_0 \subseteq E_1 \subseteq \cdots \subseteq E_n = F \quad (4)$$

and each extension E_i/E_{i-1} has degree one or two. If the degree is 2, we write $E_i = E_{i-1}(\sqrt{a_i})$ for some $a_i \in E_{i-1}$. Thus, since each nontrivial step has degree 2, the degree $[F : \mathbb{Q}]$ equals the product of the degrees of each step, and is a power of two. $[F : \mathbb{Q}] = 2^n$. $\square$

We may rephrase this: α is constructible over $\mathbb{Q} \iff \mathbb{Q} = K_0 \subseteq K_1 \subseteq K_2 \subseteq \dots \subseteq K_m = \mathbb{Q}(\alpha)$ with $[K_{i+1} : K_i] = 2$.

Corollary 4. *A regular n -gon is constructible $\iff \zeta_n$ is constructible.*

Proof. $\Leftarrow$: If you can construct the complex number $\zeta_n = \cos(\frac{2\pi}{n}) + i\sin(\frac{2\pi}{n})$ in the plane, then you can place the center at the origin and mark the point ζ_n on the unit circle. Rotating that point by successive powers ζ_n^k (or just repeating the compass-and-ruler step of copying the chord/angle) gives all n vertices. So you can construct the whole regular n -gon. $\checkmark$

$\Rightarrow$: Assume we can construct a given regular n -gon. Then given a segment, we can construct a polygon similar to a regular n -gon. Equivalently, we can construct the polygon's center and one vertex. Place the polygon so its center is the origin and the vertex is on the unit circle. So the coordinates of that vertex are $(\cos(\frac{2\pi}{n}), \sin(\frac{2\pi}{n}))$. $\checkmark$

□

Theorem 5. $[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \phi(n)$

Proof. Let $K = \mathbb{Q}(\zeta_n)$. First note that if $\sigma : K \hookrightarrow \mathbb{C}$ is a $\mathbb{Q}$ -embedding then $\sigma(\zeta_n)^n = \sigma(\zeta_n^n) = \sigma(1) = 1$. So $\sigma(\zeta_n)$ is an n -th root of unity. Since σ is injective, $|\sigma(\zeta_n)| = |\zeta_n| = 1$. So $\sigma(\zeta_n)$ is too a primitive n -th root of unity. So every map sends ζ_n to some ζ_n^a with $(a, n) = 1$. Since the map is completely determined by where it sends ζ_n , the number of distinct $\mathbb{Q}$ -embeddings $K \hookrightarrow \mathbb{C}$ is at most the number of primitive n -th roots of unity which is just $\phi(n)$.

Let a be such that $(a, n) = 1$. Define the map:

$$\sigma_a : \mathbb{Q}(\zeta_n) \rightarrow \mathbb{C}, \quad \sigma_a \left(\sum_{k=0}^m c_k \zeta_n^k \right) = \sum_{k=0}^m c_k \zeta_n^{ak} \quad (5)$$

for $c_k \in \mathbb{Q}$. This map is a field homomorphism since it is multiplicative on the generators, and linear. The image ζ_n^a has the same minimal polynomial degree as ζ_n since it is primitive. So σ_a is injective and therefore an embedding. So for every unit $a \in (\mathbb{Z}/n)^\times$, we have an embedding σ_a . Different residues $a \bmod n$ with $(a, n) = 1$ give different images of ζ_n , so different embeddings. Hence, we have at least $\phi(n)$ many embeddings. So from these two observations, we may realize that the number of embeddings is precisely $\phi(n)$.

Finally, realize that the number of distinct $\mathbb{Q}$ -embeddings $K \hookrightarrow \mathbb{C} = [K : \mathbb{Q}]$. So we have that $[K : \mathbb{Q}] = \phi(n)$. □

Theorem 6. ζ_n is constructible $\iff \phi(n) = 2^m$ for some m .

Proof. $\Rightarrow$: ζ_n is constructible $\Rightarrow \mathbb{Q} = K_0 \subseteq K_1 \subseteq \dots \subseteq K_m = \mathbb{Q}(\zeta_n)$ such that $[K_{i+1} : K_i] = 2$. This implies $[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = 2^m$. But we know $[\mathbb{Q}(\zeta_n) : \mathbb{Q}] = \phi(n)$ from the above theorem. $\checkmark$

$\Leftarrow$: Since $G = \text{Gal}(\mathbb{Q}(\zeta_n) : \mathbb{Q})$ is a finite abelian group of order $\phi(n) = 2^m$, there exists a chain of subgroups:

$$\langle e \rangle = G_m \leq G_{m-1} \leq \dots \leq G_0 = G \quad (6)$$

such that $[G_{i-1} : G_i] = 2$. Setting $K_i = \mathbb{Q}(\zeta_n)^{G_i}$ shows that ζ_n is constructible $\checkmark$ □

Remark 4. So to find when a regular n -gon is constructible, we have reduced it to finding out when $\phi(n) = 2^m$. We know that

$$\phi(n) = \phi(p_1^{a_1} \dots p_r^{a_r}) \quad (7)$$

$$= \phi(p_1^{a_1}) \dots \phi(p_r^{a_r}) \quad (8)$$

$$= (p_1^{a_1-1}) \dots (p_r^{a_r-1}) \quad (9)$$

So we've reduced even further to when is $p^j - p^{j-1}$ a power of 2? If $j \geq 2$, $p | (p^j - p^{j-1}) \Rightarrow p = 2$. If $j = 1$, then we need $p - 1 = 2^b$ for some b . This implies $p = 2^{2^b} + 1$, a Fermat prime. Thus, a regular n -gon is constructible if and only if $n = 2^k p_1 \cdot p_2 \cdot \dots \cdot p_r$ for p_i , $i \geq 0$ distinct Fermat primes.

References

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